

SI-HEP Quantity Passport

A reversible dimensional contract for relativistic beam observation

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Status. Normative infrastructure for GO Core v0.2 and the LHC beam module. The passport fixes the meaning of $c = \hbar = 1$, gives an invertible SI-HEP chart on the mechanical sector, and defines exact accelerator-unit bridges. It does not propose a new physical law.

Abstract

Natural units suppress conversion constants in formulas but do not erase physical dimensions. For a mechanical quantity with SI dimension $L^a M^b T^\tau$, fix an energy anchor $\mathcal{E}_0 > 0$, set

$$d_E = b - a - \tau, \quad u = a + \tau, \quad v = a - 2b,$$

and define the SI representative of one natural unit by

$$\mathcal{U}_{\text{SI}}(a, b, \tau; \mathcal{E}_0) = \hbar^u c^v \mathcal{E}_0^{d_E}.$$

Then $\mathcal{N}_{\mathcal{E}_0}(Q) = Q/\mathcal{U}_{\text{SI}}(a, b, \tau; \mathcal{E}_0)$ is dimensionless and the inverse map is multiplication by the same representative. This gives a multiplicative, exponent-additive, reversible chart on the LMT sector. It does not automatically cover current, temperature, amount of substance, luminous intensity, or electromagnetic field normalization. For LHC beam kinematics the missing electromagnetic step is bypassed by the exact mixed-unit identity

$$p[\text{GeV}/c] = 0.299792458 z B\rho[\text{T m}], \quad q = ze.$$

The passport also separates the SI momentum chart $p^\mu = (E/c, \mathbf{p})$, the SI energy chart $\Pi^\mu = (E, c\mathbf{p})$, and the HEP numerical chart. Consequently the usual shorthand $E^2 = \mathbf{p}^2 + m^2$ is admitted only after every component has been represented in the same energy unit. Exact definition-based conversions, measured constants, and operational machine parameters are assigned distinct uncertainty statuses.

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1 Scope and failure mode

The blocked LHC manuscript used four locally familiar notations:

$$E [\text{TeV}], \quad p [\text{GeV}/c], \quad B\rho [\text{T m}], \quad E^2 = \mathbf{p}^2 + m^2 \quad (c = 1).$$

Each can be correct. Their untyped composition is not a quantity calculus. In particular, the last equation does not permit an SI mass in kilograms to be added to a numerical momentum in GeV/c .

Boundary 1.1 (Claim firewall). Setting $c = \hbar = 1$ is a coordinate choice on the dimension group, not an empirical statement that the constants have ceased to exist. It neither fixes an electromagnetic rationalization convention nor makes tesla an energy squared without further definitions.

The present passport covers:

- (i) the mechanical LMT sector;
- (ii) relativistic four-momentum in a declared metric signature;
- (iii) charged-particle magnetic rigidity through a mixed-unit bridge;
- (iv) RF voltage-to-energy conversion through a declared charge state;
- (v) uncertainty and provenance fields needed for reproducibility.

2 Typed quantity records

Definition 2.1 (Quantity record). A quantity record is

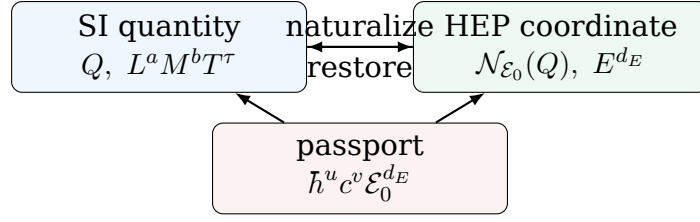
$$\mathbf{q} = (\text{id}, Q, \mathbf{d}, \text{kind}, \text{frame}, \text{context}, \text{representation}, \mathbf{U}),$$

where Q is the physical quantity, $\mathbf{d}$ is its seven-component SI dimension vector, kind is its semantic type, frame is a reference-frame label. The context obeys

$$\text{context} \in \{\text{SI}, \text{HEP-natural}, \text{accelerator-mixed}\}.$$

The field representation specifies which powers of c and $\hbar$ have been absorbed, and $\mathbf{U}$ contains exactness, uncertainty, and provenance.

Equality and addition are permitted only for records with compatible physical dimension, semantic type, frame law, and representation. Multiplication combines dimension vectors and representations. A context change is an explicit map; it is never inferred from a unit label alone.



3 Mechanical naturalization theorem

Let the admissible dimension lattice be

$$\mathcal{D}_{\text{mech}} = \{(a, b, \tau, 0, 0, 0, 0) : a, b, \tau \in \mathbb{Q}\}.$$

The last four exponents must vanish. Fractional exponents are allowed when a positive branch is part of the quantity definition.

Theorem 3.1 (Unique $c, \hbar$ reduction to an energy dimension). *For $D = L^a M^b T^\tau$, there is a unique triple $(d_E, u, v) \in \mathbb{Q}^3$ such that*

$$D = [\hbar]^u [c]^v [E]^{d_E}.$$

It is

$$\boxed{d_E = b - a - \tau, \quad u = a + \tau, \quad v = a - 2b.}$$

Proof. Using $[E] = ML^2T^{-2}$, $[\hbar] = ML^2T^{-1}$, and $[c] = LT^{-1}$, equality of the M, L, T exponents gives

$$d_E + u = b, \quad 2d_E + 2u + v = a, \quad -2d_E - u - v = \tau.$$

This linear system has determinant 1, hence a unique solution. Solving yields the displayed exponents. $\square$

Definition 3.2 (Natural coordinate and inverse). Fix an energy anchor $\mathcal{E}_0$, here 1 GeV. Define

$$\mathcal{U}_{\text{SI}}(D; \mathcal{E}_0) = \hbar^{a+\tau} c^{a-2b} \mathcal{E}_0^{b-a-\tau}, \quad \mathcal{N}_{\mathcal{E}_0}(Q) = \frac{Q}{\mathcal{U}_{\text{SI}}(D; \mathcal{E}_0)}.$$

The inverse is

$$Q = \mathcal{N}_{\mathcal{E}_0}(Q) \mathcal{U}_{\text{SI}}(D; \mathcal{E}_0).$$

Proposition 3.3 (Algebraic consistency). *For admissible positive quantities Q_1, Q_2 and rational r ,*

$$\mathcal{N}_{\mathcal{E}_0}(Q_1 Q_2) = \mathcal{N}_{\mathcal{E}_0}(Q_1) \mathcal{N}_{\mathcal{E}_0}(Q_2), \quad \mathcal{N}_{\mathcal{E}_0}(Q_1^r) = \mathcal{N}_{\mathcal{E}_0}(Q_1)^r.$$

For quantities of the same complete type, $\mathcal{N}_{\mathcal{E}_0}(Q_1 + Q_2) = \mathcal{N}_{\mathcal{E}_0}(Q_1) + \mathcal{N}_{\mathcal{E}_0}(Q_2)$.

Proof. The exponent maps $D \mapsto (d_E, u, v)$ are linear, so the SI representatives multiply under dimension addition. Division by the representative proves the product and power laws. The addition law holds only when both terms share the same representative. $\square$

If the anchor changes from $\mathcal{E}_0$ to $\mathcal{E}'_0 = \lambda \mathcal{E}_0$, then

$$\mathcal{N}_{\mathcal{E}'_0}(Q) = \lambda^{-d_E} \mathcal{N}_{\mathcal{E}_0}(Q).$$

The physical quantity is unchanged; its numerical coordinate is covariant with its energy dimension.

4 Standard conversion rows

The defining values

$$c = 299\,792\,458 \text{ m s}^{-1}, \quad h = 6.62607015 \times 10^{-34} \text{ J s}, \quad e = 1.602176634 \times 10^{-19} \text{ C}$$

are exact in the SI. Hence

$$1 \text{ eV} = e(1 \text{ V}) = 1.602176634 \times 10^{-19} \text{ J}, \quad 1 \text{ GeV} = 1.602176634 \times 10^{-10} \text{ J}$$

are exact. The displayed decimal for $\hbar = h/(2\pi)$ is rounded, although $\hbar$ is an exactly defined quantity.

Quantity	HEP unit	SI representative	Exactness
energy	GeV	$1.60217663400000 \times 10^{-10} \text{ J}$	definition-exact; decimal rounded
mass	GeV/c^2	$1.78266192162790 \times 10^{-27} \text{ kg}$	definition-exact; decimal rounded
momentum	GeV/c	$5.34428599267831 \times 10^{-19} \text{ kg m s}^{-1}$	definition-exact; decimal rounded
length	GeV^{-1}	$1.97326980459302 \times 10^{-16} \text{ m}$	definition-exact; decimal rounded
duration	GeV^{-1}	$6.58211956950907 \times 10^{-25} \text{ s}$	definition-exact; decimal rounded
area / cross section	GeV^{-2}	$3.89379372171859 \times 10^{-32} \text{ m}^2$	definition-exact; decimal rounded

Useful equivalent forms are

$$\hbar c = 197.326980459302 \text{ MeV fm}, \quad 1 \text{ GeV}^{-2} = 0.389379372172 \text{ mb}.$$

The time row uses angular frequency:

$$E = \hbar\omega, \quad 1 \text{ GeV} \longleftrightarrow 1.519267447878626 \times 10^{24} \text{ rad s}^{-1}.$$

For ordinary frequency one must instead use $E = h\nu$, giving

$$1 \text{ GeV} \longleftrightarrow 2.417989242084918 \times 10^{23} \text{ Hz}.$$

The factor 2π is part of the passport and cannot be discarded.

5 Four-momentum and invariant-mass passports

Adopt metric signature $\eta = \text{diag}(1, -1, -1, -1)$. There are three representations of the same four-momentum.

Representation	Definition	Mass shell
SI momentum chart	$p_{\text{SI}}^\mu = (E/c, \mathbf{p})$	$p_{\text{SI},\mu} p_{\text{SI}}^\mu = m^2 c^2$
SI energy chart	$\Pi_{\text{SI}}^\mu = c p_{\text{SI}}^\mu = (E, c\mathbf{p})$	$\Pi_{\text{SI},\mu} \Pi_{\text{SI}}^\mu = (mc^2)^2$
HEP numerical chart	$P_{\text{HEP}}^\mu = \Pi_{\text{SI}}^\mu / \mathcal{E}_0$	$P_{\text{HEP},\mu} P_{\text{HEP}}^\mu = (mc^2 / \mathcal{E}_0)^2$

Thus the collider shorthand

$$m^2 = E^2 - \|\mathbf{p}\|^2$$

is legal only when m means the energy-valued mass mc^2 , $\mathbf{p}$ means $c\mathbf{p}_{\text{SI}}$, and all three numerical entries use the same energy anchor. The passport-preferred notation is

$$\mu := mc^2, \quad \boldsymbol{\pi} := c\mathbf{p}_{\text{SI}}, \quad \mu^2 = E^2 - \|\boldsymbol{\pi}\|^2.$$

For a reconstructed system,

$$\mu_{\text{inv}}^2 = \left(\sum_i E_i \right)^2 - \left\| \sum_i \boldsymbol{\pi}_i \right\|^2.$$

Equivalently in SI momentum coordinates,

$$m_{\text{inv}}^2 c^4 = \left(\sum_i E_i \right)^2 - c^2 \left\| \sum_i \mathbf{p}_i \right\|^2.$$

These are the same equation under the explicit map $\boldsymbol{\pi}_i = c\mathbf{p}_i$.

6 Exact accelerator bridges

6.1 Magnetic rigidity

For transverse motion in a magnetic field,

$$|q| B\rho = |\mathbf{p}|_{\text{SI}}.$$

Let $q = ze$, with signed dimensionless charge state z . Write $e_0 = e/(1\text{ C})$ and $c_0 = c/(1\text{ m s}^{-1})$ for the exact numerical values in the declared SI units. Then

$$q = ze_0 \text{ C}, \quad 1 \text{ GeV} = 10^9 e_0 \text{ J}, \quad 1 \text{ GeV}/c = \frac{10^9 e_0}{c_0} \text{ kg m s}^{-1},$$

division by the momentum unit gives

$$\boxed{p[\text{GeV}/c] = 0.299792458 z B\rho[\text{T m}].}$$

The coefficient is exact for the declared units because

$$0.299792458 = \frac{c[\text{m s}^{-1}]}{10^9}.$$

Signs encode bending orientation; magnitudes use $|z|$.

Boundary 6.1 (Geometry boundary). The dipole bending radius is not $C/(2\pi)$ for a ring with straight sections. A circumference may determine the revolution frequency, but it cannot be substituted for ρ in $B\rho$ without a lattice model.

6.2 RF voltage

For a particle crossing an idealized RF gap at phase ϕ ,

$$\Delta E[\text{J}] = q[\text{C}] V[\text{V}] \sin \phi.$$

With $q = ze$, the exact unit identity is

$$\boxed{\Delta E[\text{eV}] = z V[\text{V}] \sin \phi, \quad \Delta E[\text{GeV}] = 10^{-9} z V[\text{V}] \sin \phi.}$$

The conversion is exact. The single-gap sinusoidal model is an approximation to a real cavity and has a distinct model-error status.

7 Electromagnetic normalization boundary

The mechanical theorem requires zero exponents of I, Θ, N, J . It therefore does not naturalize coulombs, amperes, tesla, or volts by itself. In a rationalized Heaviside-Lorentz field convention one often writes

$$e_{\text{HL}} = \sqrt{4\pi\alpha},$$

whereas other conventions distribute factors of 4π differently between fields and couplings. The statements

$$c = \hbar = 1, \quad B \sim E^2$$

do not select among these conventions.

The normative LHC policy is therefore:

- (a) retain the Lorentz force and B in SI;
- (b) encode charge as $q = ze$;
- (c) cross to HEP momentum only through the rigidity bridge;
- (d) do not register a standalone tesla-to-GeV² conversion.

This is a deliberate restriction, not a missing calculation.

8 Exactness and uncertainty

Status	Examples	Rule
Definition-exact	$c, h, e, \text{eV} \leftrightarrow \text{J}$, rigidity coefficient	Adds no physical uncertainty; printed decimals may be rounded.
Measured constant	proton mass energy	Retain the source value, standard uncertainty, edition, and covariance when relevant.
Operational reference	LHC beam energy, bunch population	Record epoch, machine mode, nominal/measured status, and precision.
Rounded geometry	public circumference 26659 m	Do not report derived revolution frequency as more accurate than the input geometry.
Model-dependent	single-gap RF gain, ideal linear optics	Separate model discrepancy from unit conversion.

For a differentiable derived vector $y = f(x)$, the first-order covariance rule is

$$\Sigma_y = J_f \Sigma_x J_f^T.$$

Exact conversion constants enter J_f but contribute no independent variance. Correlated inputs must not be propagated as independent errors.

9 Normative machine contract

A valid conversion event must contain:

(quantity_id, semantic_kind, d_{SI} , frame,
source_context, source_unit, target_context, target_unit,
absorbed_constants, $\mathcal{E}_0$, representation, exactness, uncertainty, provenance).

The conversion is rejected if:

- a residual I, Θ, N, J dimension crosses into the mechanical HEP chart without an extension contract;
- m, p , and E are combined without a common four-momentum representation;
- a field is converted without an electromagnetic convention;
- an operational LHC value lacks an epoch or mode;
- a rounded public input is marked exact.

10 Claim audit

Statement	Status	Reason
Mechanical exponent map	theorem	linear dimension algebra
Round-trip conversion	theorem	inverse by construction
Rigidity coefficient 0.299792458	exact identity	SI definitions and declared units
$\mu^2 = E^2 - \pi^2$	standard result	Lorentz metric and representation choice
Run 3 6.8 TeV reference	operational datum	CERN epoch-dependent value
Completeness for all SI quantities	rejected	electromagnetic and other residual dimensions excluded
New physical prediction	not claimed	passport is infrastructure

References

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